

Summary of Current Model Verification Study

Goal

The goal of this round was to find a concrete fermionic model that can support the perturbative SDP improvement framework in the draft. In particular, we wanted to verify three points:

1. exactness of the reduced SDP at a reference Hamiltonian H_0 ;
2. first-order response matching between the reduced SDP and the physical ground state at H_0 ;
3. a nontrivial lower bound

$$\|\delta m_S(\theta)\| \geq c_{\text{mis}}|\theta|$$

for selected moments after a weak perturbation $H(\theta) = H_0 + \theta V$.

We tested two spinful Hubbard-type candidates, both based on an isolated Hubbard-dimer reference point.

Common Setup And Notation

- Lattice: an open one-dimensional chain with L sites, L even.
- Filling: half filling, $N_e = L$ electrons.
- Strong dimers: $D_r = (2r - 1, 2r)$, $r = 1, \dots, L/2$.
- Weak inter-dimer bonds: $B_r = (2r, 2r + 1)$, $r = 1, \dots, L/2 - 1$.
- Operators:
 - $c_{i,\sigma}, c_{i,\sigma}^\dagger$: fermion annihilation/creation operators at site i with spin $\sigma \in \{\uparrow, \downarrow\}$.
 - $n_{i,\sigma} = c_{i,\sigma}^\dagger c_{i,\sigma}$, and $n_i = n_{i,\uparrow} + n_{i,\downarrow}$.
 - $D_i = n_{i,\uparrow} n_{i,\downarrow}$, the onsite double occupancy.
 - $K_{ij} = \sum_\sigma (c_{i,\sigma}^\dagger c_{j,\sigma} + c_{j,\sigma}^\dagger c_{i,\sigma})$, the bond kinetic observable.
 - $S_i^z = (n_{i,\uparrow} - n_{i,\downarrow})/2$, $S_i^+ = c_{i,\uparrow}^\dagger c_{i,\downarrow}$, and $S_i^- = c_{i,\downarrow}^\dagger c_{i,\uparrow}$.
 - *h.c.* denotes the Hermitian conjugate.

The common reference Hamiltonian is the isolated-dimer spinful Hubbard model with $t_d = 1$:

$$H_0 = \sum_{r=1}^{L/2} [-K_{2r-1,2r} + U(D_{2r-1} + D_{2r})].$$

For response matching, fields are coupled to the selected moments $M_1, \dots, M_s$ by

$$H(\theta, h) = H_0 + \theta V - \sum_{a=1}^s h_a M_a.$$

Reported Quantities

- Reference energy error: $|E_{\text{SDP}}(H_0) - E_{\text{phys}}(H_0)|$.
- Selected-moment error: $\|m_S^{\text{SDP}}(0) - m_S^{\text{phys}}(0)\|$.
- Response-matching error: $\|D_h m_S^{\text{SDP}}(0) - D_h m_S^{\text{phys}}(0)\|$, the norm of the difference between the SDP and physical first-order response matrices.
- Mismatch ratio: $\|\delta m_S(\theta)\|/|\theta|$, where $\delta m_S(\theta) = m_S^{\text{SDP}}(\theta) - m_S^{\text{phys}}(\theta)$. This ratio is used to estimate c_{mis} .
- Solver: all reported SDP values below were computed with MOSEK.

Model 1: Standard Dimerized Spinful Hubbard Chain

Model. The first model is the standard alternating-hopping spinful Hubbard chain. It uses the common reference Hamiltonian H_0 above and the weak inter-dimer hopping perturbation

$$V_{\text{hop}} = - \sum_{r=1}^{L/2-1} K_{2r,2r+1}.$$

The reported scan used $L = 4$, $U = 2$, $t_d = 1$, and $\theta = \pm 0.005, \pm 0.01$.

The selected moment families tested were:

- Dimer-local bond kinetic moments: $M_r = K_{2r-1,2r}$.
- Dimer-local double occupancy moments: $M_r = D_{2r-1} + D_{2r}$, and also the site-resolved versions D_{2r-1} , D_{2r} .
- Dimer-local spin moments: $S_{2r-1}^z S_{2r}^z$, and $\mathbf{S}_{2r-1} \cdot \mathbf{S}_{2r} = S_{2r-1}^z S_{2r}^z + \frac{1}{2}(S_{2r-1}^+ S_{2r}^- + S_{2r-1}^- S_{2r}^+)$.
- Dimer-local pair-hopping moments: $P_r = c_{2r-1,\uparrow}^\dagger c_{2r-1,\downarrow}^\dagger c_{2r,\downarrow} c_{2r,\uparrow} + h.c.$.
- Dimer-local singlet/triplet projectors: $\Pi_{s,r} = |s_r\rangle\langle s_r|$ and $\Pi_{t0,r} = |t_r^0\rangle\langle t_r^0|$, where $|s_r\rangle = (|\uparrow, \downarrow\rangle - |\downarrow, \uparrow\rangle)/\sqrt{2}$ and $|t_r^0\rangle = (|\uparrow, \downarrow\rangle + |\downarrow, \uparrow\rangle)/\sqrt{2}$ on dimer D_r .
- Inter-dimer bond kinetic moments: $M_r = K_{2r,2r+1}$ on the weak bonds B_r .

Verification results.

1. **Reference exactness.** With MOSEK, the reference energy error was around 10^{-9} .
2. **First-order response matching.** Dimer-local selected moments matched well at H_0 . Inter-dimer bond moments gave a large response-matching error.
3. **Mismatch lower bound.** For all dimer-local observables tested, including bond kinetic, double occupancy, spin correlations, pair hopping, and singlet/triplet projectors, the mismatch scaled quadratically:

$$\|\delta m_S(\theta)\| \sim C\theta^2,$$

rather than linearly in $|\theta|$. Inter-dimer bond moments did show a strong linear mismatch, while response matching failed for those moments.

Model 2: Density-Perturbed Extended Spinful Hubbard Dimer Chain

Model. The second model uses the same isolated Hubbard-dimer reference H_0 , but replaces weak hopping by an inter-dimer density-density perturbation:

$$V_{\text{dens}} = \sum_{r=1}^{L/2-1} n_{2r} n_{2r+1}.$$

The selected moments are the strong-dimer bond kinetic observables:

$$M_r = K_{2r-1,2r}, \quad r = 1, \dots, L/2.$$

The reported checks used $U = 4$, $t_d = 1$, and $L = 4, 6$.

Verification results.

1. **Reference exactness.** With MOSEK, the reference point was exact to high precision.
 - $L = 4, U = 4$: reference energy error about 5.7×10^{-12} ; selected-moment error about 1.0×10^{-7} .
 - $L = 6, U = 4$: reference energy error about 8.2×10^{-13} ; selected-moment error about 1.0×10^{-9} .
2. **First-order response matching.** The SDP and physical response matrices had small response-matching errors.
 - $L = 4, U = 4$: response-matching error about 3.0×10^{-5} .
 - $L = 6, U = 4$: response-matching error about 3.5×10^{-6} .
3. **Mismatch lower bound.** The selected bond-kinetic moments showed a stable linear mismatch after the density perturbation.
 - $L = 4, U = 4$: $\|\delta m_S(\theta)\|/|\theta| \approx 0.125$ for $|\theta| \leq 0.02$, corresponding to $c_{\text{mis}} \approx 0.12$ on the tested range.
 - $L = 6, U = 4$: $\|\delta m_S(\theta)\|/|\theta| \approx 0.215\text{--}0.216$ for $|\theta| = 0.005, 0.01$, corresponding to $c_{\text{mis}} \approx 0.21$ on the tested range.

Observed Status

For the standard hopping perturbation, the reference exactness error was small, the dimer-local response check passed numerically, and the dimer-local mismatch was quadratic in θ in the tested cases.

For the density-density perturbation, the tested $L = 4$ and $L = 6$ cases both gave small reference errors, small response-matching errors, and a linear mismatch ratio bounded away from zero on the tested θ -range.